

Anchor Padding for Minimum ℓ_q -Norm Load Balancing: Partial Progress in the Intermediate-Machine Regime

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Abstract

Status: partial progress, not a complete solution. Fix a constant $q > 1$ and consider identical-machine scheduling with objective $\sum_{i=1}^m C_i^q$. Chen, Tao, and Verschae asked for tight PTAS running times when

$$m = \Theta((1/\varepsilon)^\theta), \quad \theta \in (1/2, 1].$$

We prove a lower bound throughout the open interval $\theta \in (1/2, 1)$. The main device is an exact padding operation. If an instance on m_0 machines has total processing volume $m_0 Q$, then adding R machines together with R jobs of size Q increases the optimum by exactly RQ^q . Strict convexity forces every added job to occupy a machine by itself, so the hard core of the instance is unchanged.

Applying this operation to the ETH reduction of Chen–Tao–Verschae gives the following. For every fixed $\theta \in (1/2, 1)$ and every fixed $\delta \in (0, 1 - \theta)$, ETH rules out a $(1 + \varepsilon)$ -approximation whose running time is

$$2^{O((1/\varepsilon)^{1-\theta-\delta})} + \langle I \rangle^{O(1)}$$

even on a hard family satisfying $m = \Theta((1/\varepsilon)^\theta)$. In particular, there is no FPTAS on this promise family under ETH. The proof does not give a matching upper bound and does not treat the endpoint $\theta = 1$; hence it is a substantive partial resolution rather than a solution of the full open problem.

1 Introduction

We study the scheduling problem

$$P \parallel \sum_i C_i^q, \quad q > 1,$$

on identical parallel machines. An instance consists of m machines and jobs with positive processing times p_j . If C_i denotes the total load assigned to machine i , the objective is to minimize

$$\sum_{i=1}^m C_i^q.$$

The exponent q is fixed and is not part of the input.

Chen, Tao, and Verschae [1] established a PTAS with running time $2^{\tilde{O}(\sqrt{1/\varepsilon})} + n^{O(1)}$ and an essentially matching ETH lower bound in the regime where the number of machines is approximately $\sqrt{1/\varepsilon}$. They also showed that substantially faster algorithms are possible in other ranges of m , and explicitly left open the determination of tight running times when

$$m = \Theta((1/\varepsilon)^\theta), \quad \theta \in (1/2, 1].$$

This note makes progress on the lower-bound side of that question. The key observation is that the hard instances in [1] have a known average load Q . One may enlarge the number of machines without allowing the hard jobs to spread over the new machines by pairing every new machine with one new job of size exactly Q . A two-stage exchange argument shows that, in every optimum, the new jobs are singleton jobs and the original jobs remain on precisely the original number of machines. Thus the optimum is shifted by a known additive constant, and the original additive YES/NO gap is preserved exactly.

The resulting lower bound has a particularly transparent parameterization. For the padded hard family,

$$(\varepsilon m)^{-1} = N^{1+o(1)},$$

where N is the number of variables in the source gap-SAT instance. Calibrating the number of added anchors gives $m = \Theta((1/\varepsilon)^\theta)$ and yields an exponent $1 - \theta$ in the ETH lower bound.

What is and is not resolved. For every fixed $\theta \in (1/2, 1)$, we obtain a nontrivial ETH lower bound and exclude an FPTAS on the corresponding promise family. We do not obtain a matching algorithm, so the tight PTAS running time remains undetermined. Moreover, the construction necessarily has $\varepsilon m = o(1)$ and therefore does not reach $\theta = 1$. The full open problem is consequently not solved.

Computational convention. We take q and the other constants appearing below to be fixed computable reals. Processing times are rational and encoded in binary. When q is not an integer, objective values can be evaluated to the polynomially many bits needed to distinguish the two thresholds; this standard numerical point is discussed in Section 7.

2 The hard core extracted from Chen–Tao–Verschae

We first isolate the quantitative properties of the reduction in [1] that are needed by the padding argument. Throughout this section, N denotes the number of variables in the source instance, not the number of scheduling jobs.

Proposition 2.1 (Hard core from Chen–Tao–Verschae). *Fix $q > 1$. There exist constants $0 < a < b$ and a polynomial-time reduction from an ETH-hard promise problem on N variables to a scheduling instance $\mathcal{I}_N$ with $m_{0,N}$ machines and a positive target load Q_N such that, for all sufficiently large N ,*

$$m_{0,N} = N^{1+o(1)}, \tag{1}$$

$$Q_N = N^{1+o(1)}, \tag{2}$$

$$\sum_{j \in \mathcal{I}_N} p_j = m_{0,N} Q_N, \tag{3}$$

and the following promise holds:

$$\text{YES: } \text{OPT}(\mathcal{I}_N) \leq m_{0,N} Q_N^q + a N Q_N^{q-2}, \tag{4}$$

$$\text{NO: } \text{OPT}(\mathcal{I}_N) \geq m_{0,N} Q_N^q + b N Q_N^{q-2}. \tag{5}$$

The encoding length of $\mathcal{I}_N$ is $N^{O(1)}$. Assuming ETH, for every constant $\eta > 0$ there is no $2^{O(N^{1-\eta})}$ -time algorithm that distinguishes the two cases.

Proof. We record the constants because the strict separation is important for the padding argument. Lemma 4 of [1] supplies a constant $\beta \in (0, 1)$ and an ETH-hard promise problem. Fix a sufficiently small constant $\xi > 0$ satisfying

$$25\xi < 1 - \beta.$$

The reduction in [1] produces

$$m_{0,N} = 2\gamma N + 8N, \quad Q_N = 10^{14}\sigma_{\max},$$

where $\gamma = N^{o(1)}$ and $\sigma_{\max} = N^{1+o(1)}$. Consequently, $m_{0,N} = N^{1+o(1)}$ and $Q_N = N^{1+o(1)}$. The construction has total processing volume exactly $m_{0,N}Q_N$.

In the completeness case, the source instance has at most $4\xi N/3$ unsatisfied clauses. Lemma 22 of [1] gives

$$\text{OPT}(\mathcal{I}_N) \leq m_{0,N}Q_N^q + uNQ_N^{q-2} + o(NQ_N^{q-2}), \quad (6)$$

where

$$u = \frac{2\xi q(q-1)}{3}.$$

In the soundness case, every assignment leaves at least $4(1 - \beta - \xi)N/3$ clauses unsatisfied. Lemma 23 of [1] gives

$$\text{OPT}(\mathcal{I}_N) \geq m_{0,N}Q_N^q + vNQ_N^{q-2} + o(NQ_N^{q-2}), \quad (7)$$

where

$$v = \frac{q(q-1)(1-\beta-\xi)}{36}.$$

The leading constants are strictly separated, because

$$v - u = \frac{q(q-1)}{36}(1 - \beta - 25\xi) > 0.$$

Choose fixed rational constants a, b satisfying

$$u < a < b < v.$$

Since the two remainder terms in (6)–(7) are $o(NQ_N^{q-2})$, inequalities (4) and (5) hold for all sufficiently large N .

The stated encoding bound and the ETH lower bound are exactly the size and hardness properties of the source reduction in [1]. A constant number of exceptional small values of N has no effect on the asymptotic claim. $\square$

Remark 2.2. The use of rational a and b is only for convenient encoding of the approximation parameter later. Their precise values are irrelevant; only the fixed positive gap $b - a$ matters.

3 Exact anchor padding

The padding lemma is independent of the internal structure of the hard instance. It uses only the total processing volume and strict convexity.

Lemma 3.1 (Strict convex transfer). *Let $q > 1$, let $p > 0$, and let $a, b \geq 0$ satisfy $a - p > b$. Then*

$$a^q + b^q > (a - p)^q + (b + p)^q.$$

Proof. For fixed $p > 0$, define

$$h_p(t) = (t + p)^q - t^q, \quad t \geq 0.$$

Since

$$h'_p(t) = q((t + p)^{q-1} - t^{q-1}) > 0,$$

the function h_p is strictly increasing. Hence

$$a^q - (a - p)^q = h_p(a - p) > h_p(b) = (b + p)^q - b^q,$$

which is equivalent to the claimed inequality. $\square$

Lemma 3.2 (Exact anchor padding). *Let $\mathcal{I}$ be a scheduling instance on m_0 identical machines, and suppose its total processing volume is m_0Q for some $Q > 0$. For an integer $R \geq 0$, form $\mathcal{I}^{+R}$ by adding R machines and R new jobs, each of processing time Q . Then*

$$\text{OPT}(\mathcal{I}^{+R}) = RQ^q + \text{OPT}(\mathcal{I}).$$

More strongly, every noncanonical schedule of $\mathcal{I}^{+R}$ admits a strictly cost-decreasing move, where a schedule is called canonical if each new job is alone on a machine and all original jobs are assigned to the remaining m_0 machines.

Proof. Call the R new jobs *anchors*. The total processing volume of the padded instance is

$$m_0Q + RQ = (m_0 + R)Q,$$

so its average machine load is exactly Q .

We first eliminate collisions between anchors. Suppose that some machine contains at least two anchors. Its load is $a \geq 2Q$. Because the average load is Q , another machine has load $b < Q$. Such a machine contains no anchor, since an anchor by itself has size Q . Move one anchor from the first machine to the second. After removing the anchor, the source load is $a - Q \geq Q > b$. Lemma 3.1, applied with $p = Q$, shows that this move strictly decreases the objective. It also increases by one the number of machines occupied by anchors. Repeating finitely many times places the anchors on distinct machines.

Now suppose the anchors are on distinct machines but an original job of size p shares a machine with an anchor. There are exactly m_0 anchor-free machines. Since the original jobs have total volume m_0Q and some positive original volume is currently assigned to anchor machines, the total load on the anchor-free machines is strictly less than m_0Q . Thus one anchor-free machine has load $b < Q$. Let a be the load of the anchor machine containing the selected original job. After removing that job, the anchor remains, so $a - p \geq Q > b$. Moving the original job to the anchor-free machine strictly decreases the objective by Lemma 3.1. Each such move reduces by one the number of original jobs placed on anchor machines, so this phase also terminates.

We have shown that every noncanonical schedule can be improved strictly. Therefore every optimal schedule is canonical. Its R singleton anchors contribute exactly RQ^q , while the original jobs form a feasible schedule of $\mathcal{I}$ on the remaining m_0 machines. This proves the lower bound

$$\text{OPT}(\mathcal{I}^{+R}) \geq RQ^q + \text{OPT}(\mathcal{I}).$$

Conversely, adjoining R singleton anchors to an optimal schedule of $\mathcal{I}$ gives a padded schedule of cost $RQ^q + \text{OPT}(\mathcal{I})$. Equality follows. $\square$

Remark 3.3. Adding empty machines is not an exact padding operation: it permits the original jobs to spread over more machines and can alter the hard core. Anchors prevent this. The average load remains Q , and strict convexity forces each new machine–job pair to decouple from the original instance in every optimum.

4 The lower-bound parameter $(\varepsilon m)^{-1}$

Apply Proposition 2.1 and fix a total machine count $M_N \geq m_{0,N}$. Add

$$R_N = M_N - m_{0,N}$$

anchors of size Q_N . For readability, write $Q = Q_N$, $m_0 = m_{0,N}$, and $M = M_N$ within this section. Lemma 3.2 implies that the padded YES and NO thresholds are

$$A_N^+ = MQ^q + aNQ^{q-2}, \quad (8)$$

$$B_N^+ = MQ^q + bNQ^{q-2}. \quad (9)$$

Their additive difference is

$$G_N := B_N^+ - A_N^+ = (b - a)NQ^{q-2}. \quad (10)$$

Thus padding changes the baseline but preserves the hard gap exactly.

Define the approximation parameter

$$\varepsilon_N := \frac{G_N}{3A_N^+} = \frac{(b - a)N}{3(MQ^2 + aN)}. \quad (11)$$

The second expression follows by factoring out Q^{q-2} and shows that ε_N has polynomial encoding length. In a YES instance, a $(1 + \varepsilon_N)$ -approximation returns a schedule of value at most

$$(1 + \varepsilon_N)A_N^+ = A_N^+ + \frac{G_N}{3} < B_N^+.$$

In a NO instance, every feasible schedule has value at least B_N^+ . Therefore such an approximation distinguishes the source promise problem.

The quantity controlling the reduction is obtained exactly from (11):

$$\frac{1}{\varepsilon_N M_N} = \frac{3}{b - a} \left(\frac{Q_N^2}{N} + \frac{a}{M_N} \right). \quad (12)$$

Since $M_N \geq m_{0,N} = N^{1+o(1)}$ and $Q_N = N^{1+o(1)}$, it follows that

$$\frac{1}{\varepsilon_N M_N} = \Theta\left(\frac{Q_N^2}{N}\right) = N^{1+o(1)}. \quad (13)$$

Theorem 4.1 (Generic padded lower bound). *Fix $q > 1$ and a constant $C > 1$. Let M_N be any sequence satisfying*

$$m_{0,N} \leq M_N \leq N^C,$$

and let ε_N be defined by (11). Assuming ETH, for every fixed $\eta \in (0, 1)$ there is no $(1 + \varepsilon_N)$ -approximation on this padded family with running time

$$2^{O(((\varepsilon_N M_N)^{-1})^{1-\eta})} + \langle \mathcal{I}_N^{+R_N} \rangle^{O(1)}.$$

Proof. By (13),

$$((\varepsilon_N M_N)^{-1})^{1-\eta} = N^{1-\eta+o(1)}.$$

For all sufficiently large N , this is at most $N^{1-\eta/2}$. The padded instance has polynomial encoding length because $M_N \leq N^C$, and it can be constructed in polynomial time by explicitly listing the anchors. The hypothetical approximation, followed by the threshold test above, would therefore distinguish the source promise problem in $2^{O(N^{1-\eta/2})}$ time. This contradicts Proposition 2.1. $\square$

5 Calibration to $m = \Theta((1/\varepsilon)^\theta)$

We now choose the number of anchors so that the padded instances lie in the intermediate-machine regime posed in [1].

Theorem 5.1 (Intermediate-machine ETH lower bound). *Fix computable constants $q > 1$ and $\theta \in (1/2, 1)$. For every fixed $\delta \in (0, 1 - \theta)$, ETH rules out a $(1 + \varepsilon)$ -approximation with running time*

$$2^{O((1/\varepsilon)^{1-\theta-\delta})} + \langle I \rangle^{O(1)}$$

on all instances in a certain hard family satisfying

$$m = \Theta((1/\varepsilon)^\theta).$$

Equivalently, there exist constants $c_-, c_+ > 0$ and infinitely many hard pairs (I, ε) for which

$$c_-(1/\varepsilon)^\theta \leq m(I) \leq c_+(1/\varepsilon)^\theta.$$

Proof. Define

$$X_N := \frac{Q_N^2}{N}.$$

By Proposition 2.1, $X_N = N^{1+o(1)}$. Choose

$$M_N := \left\lceil X_N^{\theta/(1-\theta)} \right\rceil. \tag{14}$$

Because $\theta > 1/2$, the exponent $\theta/(1 - \theta)$ is strictly larger than 1. Hence

$$M_N = N^{\theta/(1-\theta)+o(1)}$$

eventually dominates $m_{0,N} = N^{1+o(1)}$. Thus the number of added anchors is nonnegative for all sufficiently large N . Since $\theta < 1$, M_N is polynomial in N .

From the exact definition (11),

$$\frac{1}{\varepsilon_N} = \frac{3}{b-a} (M_N X_N + a) = \Theta(M_N X_N).$$

Together with (14), this gives

$$\frac{1}{\varepsilon_N} = \Theta\left(X_N^{1/(1-\theta)}\right)$$

and therefore

$$(1/\varepsilon_N)^\theta = \Theta\left(X_N^{\theta/(1-\theta)}\right) = \Theta(M_N).$$

Thus the constructed family satisfies $M_N = \Theta((1/\varepsilon_N)^\theta)$ with constants independent of N . Furthermore,

$$\frac{1}{\varepsilon_N} = N^{1/(1-\theta)+o(1)}. \tag{15}$$

Suppose an approximation algorithm with the stated running time existed. On the padded family, its exponential parameter would satisfy

$$\begin{aligned} (1/\varepsilon_N)^{1-\theta-\delta} &= N^{\frac{1-\theta-\delta}{1-\theta}+o(1)} \\ &= N^{1-\delta/(1-\theta)+o(1)}. \end{aligned}$$

For all sufficiently large N , this is at most

$$N^{1-\delta/(2(1-\theta))}.$$

The polynomial part of the running time is $N^{O(1)}$ because the padded instance has polynomial size. The threshold test following (11) would therefore solve the source promise problem in $2^{O(N^{1-\rho})}$ time for the fixed constant $\rho = \delta/(2(1-\theta)) > 0$, contradicting Proposition 2.1. $\square$

Corollary 5.2 (No FPTAS on the promise family). *Fix $q > 1$ and $\theta \in (1/2, 1)$. Unless ETH fails, there is no approximation scheme running in time polynomial in both $\langle I \rangle$ and $1/\varepsilon$ on the promise family from Theorem 5.1.*

Proof. Choose any $\delta \in (0, 1-\theta)$ and set $c = 1-\theta-\delta > 0$. A running time polynomial in $1/\varepsilon$ is bounded, for all sufficiently small ε , by $2^{O((1/\varepsilon)^c)}$. This contradicts Theorem 5.1. $\square$

6 Limitations and the endpoint $\theta = 1$

Theorem 5.1 is a lower bound, not a tight characterization. The known general upper bound from [1] remains

$$2^{\tilde{O}(\sqrt{1/\varepsilon})} + n^{O(1)}.$$

For $\theta > 1/2$, the lower-bound exponent $1-\theta$ is strictly smaller than $1/2$, so a nontrivial gap remains between the known upper and lower bounds. Closing that gap requires a new algorithmic idea, a stronger hard core, or both.

The endpoint $\theta = 1$ is not reached by anchor padding. Indeed, (11) gives

$$\varepsilon_N M_N = \frac{(b-a)NM_N}{3(M_N Q_N^2 + aN)} = \Theta\left(\frac{N}{Q_N^2}\right) = N^{-1+o(1)}, \quad (16)$$

for every choice $M_N \geq m_{0,N}$. In particular, $\varepsilon_N M_N \rightarrow 0$. By contrast, the relation $M_N = \Theta(1/\varepsilon_N)$ would require $\varepsilon_N M_N = \Theta(1)$. Since the left-hand side of (16) is asymptotically independent of the number of anchors, adding any polynomial number of them cannot repair this mismatch.

Thus the present method yields substantive partial progress for every fixed $\theta \in (1/2, 1)$ but neither resolves the endpoint $\theta = 1$ nor identifies the tight PTAS running time in the interior.

7 Numerical and encoding details

The source construction uses polynomially bounded rational processing times, and the anchor size Q_N has polynomial encoding length. The number of anchors in Theorem 5.1 is polynomial in N , so the padded instance can be written explicitly in polynomial time.

The constants a and b in Proposition 2.1 were chosen rational. Consequently, the exact expression

$$\varepsilon_N = \frac{(b-a)N}{3(M_N Q_N^2 + aN)}$$

is rational and has $O(\log N)$ bits up to a fixed constant factor. If a particular implementation requires a dyadic approximation to ε_N , one may round it down by a sufficiently small constant factor; the strict one-third margin in the threshold test remains intact.

For nonintegral q , the objective of a rational schedule need not be rational. This causes no complexity issue. The separation between the YES return bound and the NO threshold is at least

$2G_N/3 = \Theta(NQ_N^{q-2})$. Since the padded instance contains only $N^{O(1)}$ machines and all loads have polynomial bit length, evaluating each term C_i^q to polynomially many bits gives total additive error less than, say, $G_N/12$. One can then compare the returned schedule with any threshold strictly between $A_N^+ + G_N/3$ and B_N^+ . This is the same standard finite-precision convention used in [1].

8 Conclusion

Anchor padding is an exact way to enlarge the machine count in a convex load balancing instance whose average load is known. Applied to the reduction of Chen–Tao–Verschae, it preserves the additive hardness gap and exposes the parameter $(\varepsilon m)^{-1}$. Calibrating the number of anchors proves that, for every fixed $\theta \in (1/2, 1)$, the exponent in the dependence on $1/\varepsilon$ cannot be reduced below $1 - \theta$ by any fixed positive power under ETH.

This answers one side of the authors’ question in the interior regime, but only partially: matching upper bounds and the endpoint $\theta = 1$ remain open within the scope of the present argument.

References

- [1] L. Chen, L. Tao, and J. Verschae, Tight running times for minimum ℓ_q -norm load balancing: beyond exponential dependencies on $1/\varepsilon$, arXiv:2107.06261, 2021. Journal version: *Mathematics of Operations Research*, DOI: [10.1287/moor.2023.0231](https://doi.org/10.1287/moor.2023.0231).
- [2] R. Impagliazzo and R. Paturi, On the complexity of k -SAT, *Journal of Computer and System Sciences* 62(2) (2001), 367–375.
- [3] R. Impagliazzo, R. Paturi, and F. Zane, Which problems have strongly exponential complexity?, *Journal of Computer and System Sciences* 63(4) (2001), 512–530.